

EIGENVALUE PROBLEMS

6.1 EIGENVALUE EQUATIONS

Many important problems in physics can be cast as equations of the generic form

$$A\psi = \lambda\psi, \quad (6.1)$$

where A is a linear operator whose domain and range is a Hilbert space, ψ is a function in the space, and λ is a constant. The operator A is known, but both ψ and λ are unknown, and the task at hand is to solve Eq. (6.1). Because the solutions to an equation of this type yield functions ψ that are unchanged by the operator (except for multiplication by a scale factor λ), they are termed **eigenvalue equations**: **Eigen** is German for “[its] own.” A function ψ that solves an eigenvalue equation is called an **eigenfunction**, and the value of λ that goes with an eigenfunction is called an **eigenvalue**.

The formal definition of an eigenvalue equation may not make its essential content totally apparent. The requirement that the operator A leaves ψ unchanged except for a scale factor constitutes a severe restriction upon ψ . The possibility that Eq. (6.1) has any solutions at all is in many cases not intuitively obvious.

To see why eigenvalue equations are common in physics, let’s cite a few examples:

1. The resonant standing waves of a vibrating string will be those in which the restoring force on the elements of the string (represented by $A\psi$) are proportional to their displacements ψ from equilibrium.
2. The angular momentum $\mathbf{L}$ and the angular velocity $\boldsymbol{\omega}$ of a rigid body are three-dimensional (3-D) vectors that are related by the equation

$$\mathbf{L} = \mathbf{I}\boldsymbol{\omega},$$

where $\mathbf{I}$ is the 3×3 moment of inertia matrix. Here the direction of $\boldsymbol{\omega}$ defines the axis of rotation, while the direction of $\mathbf{L}$ defines the axis about which angular momentum is

generated. The condition that these two axes be in the same direction (thereby defining what are known as the **principal axes** of inertia) is that $\mathbf{L} = \lambda \boldsymbol{\omega}$, where λ is a proportionality constant. Combining with the formula for $\mathbf{L}$, we obtain

$$\mathbf{I} \boldsymbol{\omega} = \lambda \boldsymbol{\omega},$$

which is an eigenvalue equation in which the operator is the matrix $\mathbf{I}$ and the eigenfunction (then usually called an **eigenvector**) is the vector $\boldsymbol{\omega}$.

3. The time-independent Schrödinger equation in quantum mechanics is an eigenvalue equation, with A the Hamiltonian operator H , ψ a wave function and $\lambda = E$ the energy of the state represented by ψ .

Basis Expansions

A powerful approach to eigenvalue problems is to express them in terms of an orthonormal basis whose members we designate φ_i , using the formulas developed in Chapter 5. Then the operator A and the function ψ are represented by a matrix A and a vector $\mathbf{c}$ whose elements are obtained, according to Eqs. (5.51) and (5.52), as the scalar products

$$a_{ij} = \langle \varphi_i | A | \varphi_j \rangle, \quad c_i = \langle \varphi_i | \psi \rangle.$$

Our original eigenvalue equation has now been reduced to a matrix equation:

$$\mathbf{A} \mathbf{c} = \lambda \mathbf{c}. \quad (6.2)$$

When an eigenvalue equation is presented in this form, we can call it a **matrix eigenvalue equation** and call the vectors $\mathbf{c}$ that solve it **eigenvectors**. As we shall see in later sections of this chapter, there is a well-developed technology for the solution of matrix eigenvalue equations, so a route always available for solving eigenvalue equations is to cast them in matrix form. Once a matrix eigenvalue problem has been solved, we can recover the eigenfunctions of the original problem from their expansion:

$$\psi = \sum_i c_i \varphi_i.$$

Sometimes, as in the moment of inertia example mentioned above, our eigenvalue problem originates as a matrix problem. Then, of course, we do not have to begin its solution process by introducing a basis and converting it into matrix form, and our solutions will be vectors that do not need to be interpreted as expansions in a basis.

Equivalence of Operator and Matrix Forms

It is important to note that we are dealing with eigenvalue equations in which the operator involved is linear and that it operates on elements of a Hilbert space. Once these conditions are met, the operator and function involved can always be expanded in a basis, leading to a matrix eigenvalue equation that is totally equivalent to our original problem. Among other things, this means that any theorems about properties of eigenvectors or eigenvalues that are developed from basis-set expansions of an eigenvalue problem must apply also to the original problem, and that solution of the matrix eigenvalue equation provides also a

solution to the original problem. These facts, plus the practical observation that we know how to solve matrix eigenvalue problems, strongly suggest that the detailed investigation of the matrix problems should be on our agenda.

When we explore matrix eigenvalue problems, we will find that certain properties of the matrix influence the nature of the solutions, and that in particular significant simplifications become available when the matrix is Hermitian. Many eigenvalue equations of interest in physics involve differential operators, so it is of importance to understand whether (or under what conditions) these operators are Hermitian. That issue is taken up in Chapter 8.

Finally, we note that the introduction of a basis-set expansion is not the only possibility for solving an eigenvalue equation. Eigenvalue equations involving differential operators can also be approached by the general methods for solving differential equations. That topic is also discussed in Chapter 8.

6.2 MATRIX EIGENVALUE PROBLEMS

While in principle the notion of an eigenvalue problem is already fully defined, we open this section with a simple example that may help to make it clearer how such problems are set up and solved.

A Preliminary Example

We consider here a simple problem of two-dimensional (2-D) motion in which a particle slides frictionlessly in an ellipsoidal basin (see Fig. 6.1). If we release the particle (initially at rest) at an arbitrary point in the basin, it will start to move downhill in the (negative) gradient direction, which in general will not aim directly at the potential minimum at the bottom of the basin. The particle's overall trajectory will then be a complicated path, as sketched in the bottom panel of Fig. 6.1. Our objective is to find the positions, if any, from which the trajectories will aim at the potential minimum, and will therefore represent simple one-dimensional oscillatory motion.

This problem is sufficiently elementary that we can analyze it without great difficulty. We take a potential of the form

$$V(x, y) = ax^2 + bxy + cy^2,$$

with parameters a, b, c in ranges that describe an ellipsoidal basin with a minimum in V at $x = y = 0$. We then calculate the x and y components of the force on our particle when at (x, y) :

$$F_x = -\frac{\partial V}{\partial x} = -2ax - by, \quad F_y = -\frac{\partial V}{\partial y} = -bx - 2cy.$$

It is pretty clear that, for most values of x and y , $F_x/F_y \neq x/y$, so the force will not be directed toward the minimum at $x = y = 0$.

To search for directions in which the force is directed toward $x = y = 0$, we begin by writing the equations for the force in matrix form:

$$\begin{pmatrix} F_x \\ F_y \end{pmatrix} = \begin{pmatrix} -2a & -b \\ -b & -2c \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}, \quad \text{or} \quad \mathbf{f} = \mathbf{H}\mathbf{r},$$

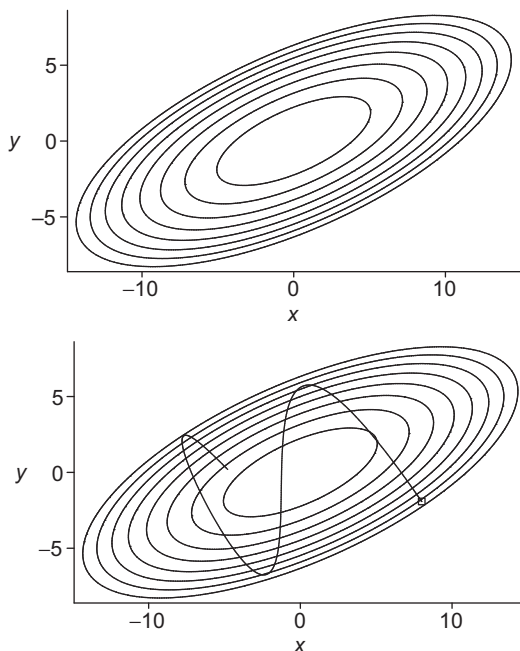


FIGURE 6.1 Top: Contour lines of basin potential $V = x^2 - \sqrt{5}xy + 3y^2$. Bottom: Trajectory of sliding particle of unit mass starting from rest at $(8.0, -1.92)$.

where $\mathbf{f}$, $\mathbf{H}$, and $\mathbf{r}$ are defined as indicated. Now the condition $F_x/F_y = x/y$ is equivalent to the statement that $\mathbf{f}$ and $\mathbf{r}$ are proportional, and therefore we can write

$$\mathbf{H}\mathbf{r} = \lambda\mathbf{r}, \quad (6.3)$$

where, as already suggested, $\mathbf{H}$ is a known matrix, while λ and $\mathbf{r}$ are to be determined. This is an eigenvalue equation, and the column vectors $\mathbf{r}$ that are its solutions are its eigenvectors, while the corresponding values of λ are its eigenvalues.

Equation (6.3) is a homogeneous linear equation system, as becomes more obvious if written as

$$(\mathbf{H} - \lambda\mathbf{I})\mathbf{r} = 0, \quad (6.4)$$

and we know from Chapter 2 that it will have the unique solution $\mathbf{r} = 0$ unless $\det(\mathbf{H} - \lambda\mathbf{I}) = 0$. However, the value of λ is at our disposal, so we can search for values of λ that cause this determinant to vanish. Proceeding symbolically, we look for λ such that

$$\det(\mathbf{H} - \lambda\mathbf{I}) = \begin{vmatrix} h_{11} - \lambda & h_{12} \\ h_{21} & h_{22} - \lambda \end{vmatrix} = 0.$$

Expanding the determinant, which is sometimes called a **secular determinant** (the name arising from early applications in celestial mechanics), we have an algebraic equation, the **secular equation**,

$$(h_{11} - \lambda)(h_{22} - \lambda) - h_{12}h_{21} = 0, \quad (6.5)$$

which can be solved for λ . The left hand side of Eq. (6.5) is also called the **characteristic polynomial** (in λ) of H , and Eq. (6.5) is for that reason also known as the **characteristic equation** of H .

Once a value of λ that solves Eq. (6.5) has been obtained, we can return to the homogeneous equation system, Eq. (6.4), and solve it for the vector $\mathbf{r}$. This can be repeated for all λ that are solutions to the secular equation, thereby giving a set of eigenvalues and the associated eigenvectors.

Example 6.2.1 2-D ELLIPSOIDAL BASIN

Let's continue with our ellipsoidal basin example, with the specific parameter values $a = 1$, $b = -\sqrt{5}$, $c = 3$. Then our matrix H has the form

$$H = \begin{pmatrix} -2 & \sqrt{5} \\ \sqrt{5} & -6 \end{pmatrix},$$

and the secular equation takes the form

$$\det(H - \lambda \mathbf{1}) = \begin{vmatrix} -2 - \lambda & \sqrt{5} \\ \sqrt{5} & -6 - \lambda \end{vmatrix} = \lambda^2 + 8\lambda + 7 = 0.$$

Since $\lambda^2 + 8\lambda + 7 = (\lambda + 1)(\lambda + 7)$, we see that the secular equation has as solutions the eigenvalues $\lambda = -1$ and $\lambda = -7$.

To get the eigenvector corresponding to $\lambda = -1$, we return to Eq. (6.4), which, written in great detail, is

$$(H - \lambda \mathbf{1})\mathbf{r} = \begin{pmatrix} -2 - (-1) & \sqrt{5} \\ \sqrt{5} & -6 - (-1) \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 & \sqrt{5} \\ \sqrt{5} & -5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0,$$

which expands into a linearly dependent pair of equations:

$$-x + \sqrt{5}y = 0$$

$$\sqrt{5}x - 5y = 0.$$

This is, of course, the intention associated with the secular equation, because if these equations were linearly independent they would inexorably lead to the solution $x = y = 0$. Instead, from either equation, we have $x = \sqrt{5}y$, so we have the eigenvalue/eigenvector pair

$$\lambda_1 = -1, \quad \mathbf{r}_1 = C \begin{pmatrix} \sqrt{5} \\ 1 \end{pmatrix},$$

where C is a constant that can assume any value. Thus, there is an infinite number of x, y pairs that define a **direction** in the 2-D space, with the magnitude of the displacement in that direction arbitrary. The arbitrariness of scale is a natural consequence of the fact that the equation system was homogeneous; any multiple of a solution of a linear homogeneous equation set will also be a solution. This eigenvector corresponds to trajectories that start from the particle at rest anywhere on the line defined by $\mathbf{r}_1$. A trajectory of this sort is illustrated in the top panel of Fig. 6.2.

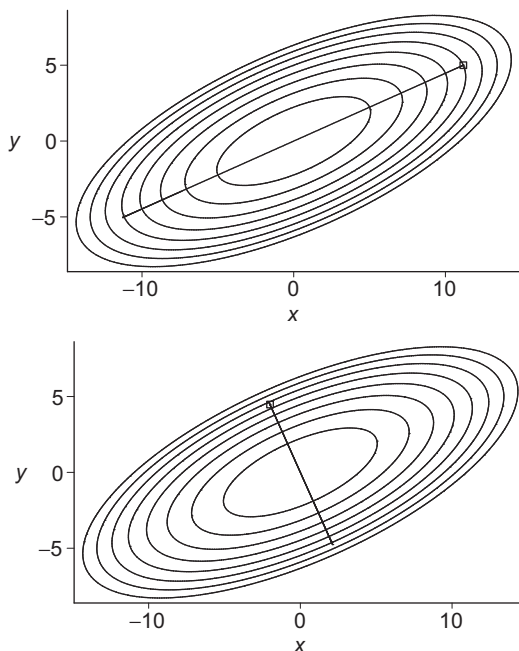


FIGURE 6.2 Trajectories starting at rest. Top: At a point on the line $x = y\sqrt{5}$. Bottom: At a point on the line $y = -x\sqrt{5}$.

We have not yet considered the possibility that $\lambda = -7$. This leads to a different eigenvector, obtained by solving

$$(\mathbf{H} - \lambda \mathbf{I})\mathbf{r} = \begin{pmatrix} -2+7 & \sqrt{5} \\ \sqrt{5} & -6+7 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 & \sqrt{5} \\ \sqrt{5} & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0,$$

corresponding to $y = -x\sqrt{5}$. This defines the eigenvalue/eigenvector pair

$$\lambda_2 = -7, \quad \mathbf{r}_2 = C' \begin{pmatrix} -1 \\ \sqrt{5} \end{pmatrix}.$$

A trajectory of this sort is shown in the bottom panel of Fig. 6.2.

We thus have two directions in which the force is directed toward the minimum, and they are mutually perpendicular: the first direction has $dy/dx = 1/\sqrt{5}$; for the second, $dy/dx = -\sqrt{5}$.

We can easily check our eigenvectors and eigenvalues. For λ_1 and $\mathbf{r}_1$,

$$\mathbf{H}\mathbf{r}_1 = \begin{pmatrix} -2 & \sqrt{5} \\ \sqrt{5} & -6 \end{pmatrix} \begin{pmatrix} C\sqrt{5} \\ C \end{pmatrix} = C \begin{pmatrix} -\sqrt{5} \\ -1 \end{pmatrix} = (-1) \begin{pmatrix} C\sqrt{5} \\ C \end{pmatrix} = \lambda_1 \mathbf{r}_1.$$

It is often useful to **normalize** eigenvectors, which we can do by choosing the constant (C or C') to make $\mathbf{r}$ of magnitude unity. In the present example,

$$\mathbf{r}_1 = \begin{pmatrix} \sqrt{5/6} \\ \sqrt{1/6} \end{pmatrix}, \quad \mathbf{r}_2 = \begin{pmatrix} -\sqrt{1/6} \\ \sqrt{5/6} \end{pmatrix}. \quad (6.6)$$

Each of these normalized eigenvectors is still arbitrary as to overall sign (or if we accept complex coefficients, as to an arbitrary complex factor of magnitude unity).

Before leaving this example, we make three further observations: (1) the number of eigenvalues was equal to the dimension of the matrix H . This is a consequence of the fundamental theorem of algebra, namely that an equation of degree n will have n roots; (2) although the secular equation was of degree 2 and quadratic equations can have complex roots, our eigenvalues were real; and (3) our two eigenvectors are orthogonal. ■

Our 2-D example is easily understood physically. The directions in which the displacement and the force are collinear are the symmetry directions of the elliptical potential field, and they are associated with different eigenvalues (the proportionality constant between position and force) because the ellipses have axes of different lengths. We have, in fact, identified the **principal axes** of our basin. With the parameters of [Example 6.2.1](#), the potential could have been written (using the normalized eigenvectors)

$$V = \frac{1}{2} \left(\frac{\sqrt{5}x + y}{\sqrt{6}} \right)^2 + \frac{7}{2} \left(\frac{x - \sqrt{5}y}{\sqrt{6}} \right)^2 = \frac{1}{2}(x')^2 + \frac{7}{2}(y')^2,$$

which shows that V divides into two quadratic terms, each dependent on a parenthesized quantity (a new coordinate) proportional to one of our eigenvectors. The new coordinates are related to the original x, y by a rotation with unitary transformation U :

$$U\mathbf{r} = \begin{pmatrix} \sqrt{5/6} & \sqrt{1/6} \\ \sqrt{1/6} & -\sqrt{5/6} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} (\sqrt{5}x + y)/\sqrt{6} \\ (x - \sqrt{5}y)/\sqrt{6} \end{pmatrix} = \begin{pmatrix} x' \\ y' \end{pmatrix}.$$

Finally, we note that when we calculate the force in the primed coordinate system, we get

$$F_{x'} = -x', \quad F_{y'} = -7y',$$

corresponding to the eigenvalues we found.

Another Eigenproblem

[Example 6.2.1](#) is not complicated enough to provide a full illustration of the matrix eigenvalue problem. Consider next the following example.

Example 6.2.2 BLOCK-DIAGONAL MATRIX

Find the eigenvalues and eigenvectors of

$$H = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}. \quad (6.7)$$

Writing the secular equation and expanding in minors using the third row, we have

$$\begin{vmatrix} -\lambda & 1 & 0 \\ 1 & -\lambda & 0 \\ 0 & 0 & 2-\lambda \end{vmatrix} = (2-\lambda) \begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = (2-\lambda)(\lambda^2 - 1) = 0. \quad (6.8)$$

We see that the eigenvalues are 2, +1, and -1.

To obtain the eigenvector corresponding to $\lambda = 2$, we examine the equation set $[\mathbf{H} - 2(\mathbf{1})]\mathbf{c} = 0$:

$$\begin{aligned} -2c_1 + c_2 &= 0, \\ c_1 - 2c_2 &= 0, \\ 0 &= 0. \end{aligned}$$

The first two equations of this set lead to $c_1 = c_2 = 0$. The third obviously conveys no information, and we are led to the conclusion that c_3 is arbitrary. Thus, at this point we have

$$\lambda_1 = 2, \quad \mathbf{c}_1 = \begin{pmatrix} 0 \\ 0 \\ C \end{pmatrix}. \quad (6.9)$$

Taking next $\lambda = +1$, our matrix equation is $[\mathbf{H} - 1(\mathbf{1})]\mathbf{c} = 0$, which is equivalent to the ordinary equations

$$\begin{aligned} -c_1 + c_2 &= 0, \\ c_1 - c_2 &= 0, \\ c_3 &= 0. \end{aligned}$$

We clearly have $c_1 = c_2$ and $c_3 = 0$, so

$$\lambda_2 = +1, \quad \mathbf{c}_2 = \begin{pmatrix} C \\ C \\ 0 \end{pmatrix}. \quad (6.10)$$

Similar operations for $\lambda = -1$ yield

$$\lambda_3 = -1, \quad \mathbf{c}_3 = \begin{pmatrix} C \\ -C \\ 0 \end{pmatrix}. \quad (6.11)$$

Collecting our results, and normalizing the eigenvectors (often useful, but not in general necessary), we have

$$\lambda_1 = 2, \quad \mathbf{c}_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad \lambda_2 = 1, \quad \mathbf{c}_2 = \begin{pmatrix} 2^{-1/2} \\ 2^{-1/2} \\ 0 \end{pmatrix}, \quad \lambda_3 = -1, \quad \mathbf{c}_3 = \begin{pmatrix} 2^{-1/2} \\ -2^{-1/2} \\ 0 \end{pmatrix}.$$

Note that because $\mathbf{H}$ was block-diagonal, with an upper-left 2×2 block and a lower-right 1×1 block, the secular equation separated into a product of the determinants for the two blocks, and its solutions corresponded to those of an individual block, with coefficients of value zero for the other block(s). Thus, $\lambda = 2$ was a solution for the 1×1 block in row/column 3, and its eigenvector involved only the coefficient c_3 . The λ values ± 1 came from the 2×2 block in rows/columns 1 and 2, with eigenvectors involving only coefficients c_1 and c_2 . ■

In the case of a 1×1 block in row/column i , we saw, for $i = 3$ in [Example 6.2.2](#), that its only element was the eigenvalue, and that the corresponding eigenvector is proportional

to $\hat{\mathbf{e}}_i$ (a unit vector whose only nonzero element is $c_i = 1$). A generalization of this observation is that if a matrix $\mathbf{H}$ is diagonal, its diagonal elements h_{ii} will be the eigenvalues λ_i , and that the eigenvectors $\mathbf{c}_i$ will be the unit vectors $\hat{\mathbf{e}}_i$.

Degeneracy

If the secular equation has a multiple root, the eigensystem is said to be **degenerate** or to exhibit **degeneracy**. Here is an example.

Example 6.2.3 DEGENERATE EIGENPROBLEM

Let's find the eigenvalues and eigenvectors of

$$\mathbf{H} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}. \quad (6.12)$$

The secular equation for this problem is

$$\begin{vmatrix} -\lambda & 0 & 1 \\ 0 & 1 - \lambda & 0 \\ 1 & 0 & -\lambda \end{vmatrix} = \lambda^2(1 - \lambda) - (1 - \lambda) = (\lambda^2 - 1)(1 - \lambda) = 0 \quad (6.13)$$

with the three roots $+1$, $+1$, and -1 . Let's consider first $\lambda = -1$. Then we have

$$c_1 + c_3 = 0,$$

$$2c_2 = 0,$$

$$c_1 + c_3 = 0.$$

Thus,

$$\lambda_1 = -1, \quad \mathbf{c}_1 = C \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}. \quad (6.14)$$

For the double root $\lambda = +1$,

$$-c_1 + c_3 = 0,$$

$$0 = 0,$$

$$c_1 - c_3 = 0.$$

Note that of the three equations, only one is now linearly independent; the double root signals **two** linear dependencies, and we have solutions for **any** values of c_1 and c_2 , with only the condition that $c_3 = c_1$. The eigenvectors for $\lambda = +1$ thus span a 2-D **manifold** (= subspace), in contrast to the trivial one-dimensional manifold characteristic of nondegenerate

solutions. The general form for these eigenvectors is

$$\lambda = +1, \quad \mathbf{c} = \begin{pmatrix} C \\ C' \\ C \end{pmatrix}. \quad (6.15)$$

It is convenient to describe the degenerate eigenspace for $\lambda = 1$ by identifying two mutually orthogonal vectors that span it. We can pick the first vector by choosing arbitrary values of C and C' (an obvious choice is to set one of these, say C' , to zero). Then, using the Gram-Schmidt process (or in this case by simple inspection), we find a second eigenvector orthogonal to the first. Here, this leads to

$$\lambda_2 = \lambda_3 = +1, \quad \mathbf{c}_2 = C \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{c}_3 = C' \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}. \quad (6.16)$$

Normalizing, our eigenvalues and eigenvectors become

$$\lambda_1 = -1, \quad \mathbf{c}_1 = \begin{pmatrix} 2^{-1/2} \\ 0 \\ -2^{-1/2} \end{pmatrix}, \quad \lambda_2 = \lambda_3 = 1, \quad \mathbf{c}_2 = \begin{pmatrix} 2^{-1/2} \\ 0 \\ 2^{-1/2} \end{pmatrix}, \quad \mathbf{c}_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

■

The eigenvalue problems we have used as examples all led to secular equations with simple solutions; realistic applications frequently involve matrices of large dimension and secular equations of high degree. The solution of matrix eigenvalue problems has been an active field in numerical analysis and very sophisticated computer programs for this purpose are now available. Discussion of the details of such programs is outside the scope of this book, but the ability to use such programs should be part of the technology available to the working physicist.

Exercises

Find the eigenvalues and corresponding normalized eigenvectors of the matrices in Exercises 6.2.1 through 6.2.14. Orthogonalize any degenerate eigenvectors.

$$6.2.1 \quad A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

ANS. $\lambda = 0, 1, 2.$

$$6.2.2 \quad A = \begin{pmatrix} 1 & \sqrt{2} & 0 \\ \sqrt{2} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

ANS. $\lambda = -1, 0, 2.$

$$6.2.3 \quad A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

ANS. $\lambda = -1, 1, 2.$

$$\mathbf{6.2.4} \quad \mathbf{A} = \begin{pmatrix} 1 & \sqrt{8} & 0 \\ \sqrt{8} & 1 & \sqrt{8} \\ 0 & \sqrt{8} & 1 \end{pmatrix}.$$

ANS. $\lambda = -3, 1, 5.$

$$\mathbf{6.2.5} \quad \mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

ANS. $\lambda = 0, 1, 2.$

$$\mathbf{6.2.6} \quad \mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \sqrt{2} \\ 0 & \sqrt{2} & 0 \end{pmatrix}.$$

ANS. $\lambda = -1, 1, 2.$

$$\mathbf{6.2.7} \quad \mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

ANS. $\lambda = -\sqrt{2}, 0, \sqrt{2}.$

$$\mathbf{6.2.8} \quad \mathbf{A} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

ANS. $\lambda = 0, 2, 2.$

$$\mathbf{6.2.9} \quad \mathbf{A} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

ANS. $\lambda = -1, -1, 2.$

$$\mathbf{6.2.10} \quad \mathbf{A} = \begin{pmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix}.$$

ANS. $\lambda = -1, 2, 2.$

$$\mathbf{6.2.11} \quad \mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

ANS. $\lambda = 0, 0, 3.$

$$\mathbf{6.2.12} \quad \mathbf{A} = \begin{pmatrix} 5 & 0 & 2 \\ 0 & 1 & 0 \\ 2 & 0 & 2 \end{pmatrix}.$$

ANS. $\lambda = 1, 1, 6.$

$$\mathbf{6.2.13} \quad \mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

ANS. $\lambda = 0, 0, 2.$

$$6.2.14 \quad A = \begin{pmatrix} 5 & 0 & \sqrt{3} \\ 0 & 3 & 0 \\ \sqrt{3} & 0 & 3 \end{pmatrix}.$$

ANS. $\lambda = 2, 3, 6.$

6.2.15 Describe the geometric properties of the surface

$$x^2 + 2xy + 2y^2 + 2yz + z^2 = 1.$$

How is it oriented in 3-D space? Is it a conic section? If so, which kind?

6.3 HERMITIAN EIGENVALUE PROBLEMS

All the illustrative problems we have thus far examined have turned out to have real eigenvalues; this was also true of all the exercises at the end of [Section 6.2](#). We also found, whenever we bothered to check, that the eigenvectors corresponding to different eigenvalues were orthogonal. The purpose of the present section is to show that these properties are consequences of the fact that all the eigenvalue problems we have considered were for Hermitian matrices.

We remind the reader that the check for Hermiticity is simple: We simply verify that H is equal to its adjoint, $H^\dagger$; if a matrix is real, this condition is simply that it be symmetric. All the matrices to which we referred are clearly Hermitian.

We now proceed to characterize the eigenvalues and eigenvectors of Hermitian matrices. Let H be a Hermitian matrix, with $\mathbf{c}_i$ and $\mathbf{c}_j$ two of its eigenvectors corresponding, respectively, to the eigenvalues λ_i and λ_j . Then, using Dirac notation,

$$H|\mathbf{c}_i\rangle = \lambda_i|\mathbf{c}_i\rangle, \quad H|\mathbf{c}_j\rangle = \lambda_j|\mathbf{c}_j\rangle. \quad (6.17)$$

Multiplying on the left the first of these by $\mathbf{c}_j^\dagger$, which in Dirac notation is $\langle\mathbf{c}_j|$, and the second by $\langle\mathbf{c}_i|$,

$$\langle\mathbf{c}_j|H|\mathbf{c}_i\rangle = \lambda_i\langle\mathbf{c}_j|\mathbf{c}_i\rangle, \quad \langle\mathbf{c}_i|H|\mathbf{c}_j\rangle = \lambda_j\langle\mathbf{c}_i|\mathbf{c}_j\rangle. \quad (6.18)$$

We next take the complex conjugate of the second of these equations, noting that $\langle\mathbf{c}_i|\mathbf{c}_j\rangle^* = \langle\mathbf{c}_j|\mathbf{c}_i\rangle$, that we must complex conjugate the occurrence of λ_j , and that

$$\langle\mathbf{c}_i|H|\mathbf{c}_j\rangle^* = \langle H\mathbf{c}_j|\mathbf{c}_i\rangle = \langle\mathbf{c}_j|H|\mathbf{c}_i\rangle. \quad (6.19)$$

Note that the first member of [Eq. \(6.19\)](#) contains the scalar product of $\mathbf{c}_i$ with $H\mathbf{c}_j$. Complex conjugating this scalar product yields the second member of that equation. The final member of the equation follows because H is Hermitian.

The complex conjugation therefore converts [Eqs. \(6.18\)](#) into

$$\langle\mathbf{c}_j|H|\mathbf{c}_i\rangle = \lambda_i\langle\mathbf{c}_j|\mathbf{c}_i\rangle, \quad \langle\mathbf{c}_j|H|\mathbf{c}_i\rangle = \lambda_j^*\langle\mathbf{c}_j|\mathbf{c}_i\rangle. \quad (6.20)$$

[Equations \(6.20\)](#) permit us to obtain two important results: First, if $i = j$, the scalar product $\langle\mathbf{c}_j|\mathbf{c}_i\rangle$ becomes $\langle\mathbf{c}_i|\mathbf{c}_i\rangle$, which is an inherently positive quantity. This means that the two equations are only consistent if $\lambda_i = \lambda_i^*$, meaning that λ_i must be real. Thus,

The eigenvalues of a Hermitian matrix are real.

Next, if $i \neq j$, combining the two equations of Eq. (6.20), and remembering that the λ_i are real,

$$(\lambda_i - \lambda_j)\langle \mathbf{c}_j | \mathbf{c}_i \rangle = 0, \quad (6.21)$$

so that either $\lambda_i = \lambda_j$ or $\langle \mathbf{c}_j | \mathbf{c}_i \rangle = 0$. This tells us that

Eigenvectors of a Hermitian matrix corresponding to different eigenvalues are orthogonal.

Note, however, that if $\lambda_i = \lambda_j$, which will occur if i and j refer to two degenerate eigenvectors, we know nothing about their orthogonality. In fact, in Example 6.2.3 we examined a pair of degenerate eigenvectors, noting that they spanned a two-dimensional manifold and were not required to be orthogonal. However, we also noted in that context that we could **choose** them to be orthogonal. Sometimes (as in Example 6.2.3), it is obvious how to choose orthogonal degenerate eigenvectors. When it is not obvious, we can start from any linearly independent set of degenerate eigenvectors and orthonormalize them by the Gram-Schmidt process.

Since the total number of eigenvectors of a Hermitian matrix is equal to its dimension, and since (whether or not there is degeneracy) we can make from them an orthonormal set of eigenvectors, we have the following important result:

*It is possible to choose the eigenvectors of a Hermitian matrix in such a way that they form an orthonormal set that spans the space of the matrix basis. This situation is often referred to by the statement, “The eigenvectors of a Hermitian matrix form a **complete set**.” This means that if the matrix is of order n , any vector of dimension n can be written as a linear combination of the orthonormal eigenvectors, with coefficients determined by the rules for orthogonal expansions.*

We close this section by reminding the reader that theorems which have been established for an arbitrary basis-set expansion of a Hermitian eigenvalue equation apply also to that eigenvalue equation in its original form. Therefore, this section has also shown that:

If H is a linear Hermitian operator on an arbitrary Hilbert space,

1. *The eigenvalues of H are real.*
2. *Eigenfunctions corresponding to different eigenvalues of H are orthogonal.*
3. *It is possible to choose the eigenfunctions of H in a way such that they form an orthonormal basis for the Hilbert space. In general, the eigenfunctions of a Hermitian operator form a complete set (i.e., a complete basis for the Hilbert space).*

6.4 HERMITIAN MATRIX DIAGONALIZATION

In Section 6.2 we observed that if a matrix is diagonal, the diagonal elements are its eigenvalues. This observation opens an alternative approach to the matrix eigenvalue problem. Given the matrix eigenvalue equation

$$H\mathbf{c} = \lambda\mathbf{c}, \quad (6.22)$$

where H is a Hermitian matrix, consider what happens if we insert unity between H and $\mathbf{c}$, as follows, with U a unitary matrix, and then left-multiply the resulting equation by U :

$$HU^{-1}U\mathbf{c} = \lambda\mathbf{c} \quad \longrightarrow \quad UHU^{-1}(U\mathbf{c}) = \lambda(U\mathbf{c}). \quad (6.23)$$

Equation (6.23) shows that our original eigenvalue equation has been converted into one in which H has been replaced by its unitary transformation (by U) and the eigenvector $\mathbf{c}$ has also been transformed by U , but the value of λ remains unchanged. We thus have the important result:

The eigenvalues of a matrix remain unchanged when the matrix is subjected to a unitary transformation.

Next, suppose that we choose U in such a way that the transformed matrix UHU^{-1} is in the eigenvector basis. While we may or may not know how to construct this U , we know that such a unitary matrix exists because the eigenvectors form a complete orthogonal set, and can be specified to be normalized. If we transform with the chosen U , the matrix UHU^{-1} will be diagonal, with the eigenvalues as diagonal elements. Moreover, the eigenvector $U\mathbf{c}$ of UHU^{-1} corresponding to the eigenvalue $\lambda_i = (UHU^{-1})_{ii}$ is $\hat{\mathbf{e}}_i$ (a column vector with all elements zero except for unity in the i th row). We may find the eigenvector $\mathbf{c}_i$ of Eq. (6.22) by solving the equation $U\mathbf{c}_i = \hat{\mathbf{e}}_i$, obtaining $\mathbf{c}_i = U^{-1}\hat{\mathbf{e}}_i$.

These observations correspond to the following:

For any Hermitian matrix H , there exists a unitary transformation U that will cause UHU^{-1} to be diagonal, with the eigenvalues of H as its diagonal elements.

This is an extremely important result. Another way of stating it is:

A Hermitian matrix can be diagonalized by a unitary transformation, with its eigenvalues as the diagonal elements.

Looking next at the i th eigenvector $U^{-1}\hat{\mathbf{e}}_i$, we have

$$\begin{pmatrix} (U^{-1})_{11} & \dots & (U^{-1})_{1i} & \dots & (U^{-1})_{1n} \\ (U^{-1})_{21} & \dots & (U^{-1})_{2i} & \dots & (U^{-1})_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ (U^{-1})_{n1} & \dots & (U^{-1})_{ni} & \dots & (U^{-1})_{nn} \end{pmatrix} \begin{pmatrix} 0 \\ \dots \\ 1 \\ \dots \\ 0 \end{pmatrix} = \begin{pmatrix} (U^{-1})_{1i} \\ (U^{-1})_{2i} \\ \dots \\ (U^{-1})_{ni} \end{pmatrix}. \quad (6.24)$$

We see that the columns of U^{-1} are the eigenvectors of H , normalized because U^{-1} is a unitary matrix. It is also clear from Eq. (6.24) that U^{-1} is not entirely unique; if its columns are permuted, all that will happen is that the order of the eigenvectors are changed, with a corresponding permutation of the diagonal elements of the diagonal matrix UHU^{-1} . Summarizing,

If a unitary matrix U is such that, for a Hermitian matrix H , UHU^{-1} is diagonal, the normalized eigenvector of H corresponding to the eigenvalue $(UHU^{-1})_{ii}$ will be the i th column of U^{-1} .

If H is not degenerate, U^{-1} (and also U) will be unique except for a possible permutation of the columns of U^{-1} (and a corresponding permutation of the rows of U). However, if H is degenerate (has a repeated eigenvalue), then the columns of U^{-1} corresponding to the same

eigenvalue can be transformed among themselves, thereby giving additional flexibility to U and U^{-1} .

Finally, calling on the fact that both the determinant and the trace of a matrix are unchanged when the matrix is subjected to a unitary transformation (shown in Section 5.7), we see that the determinant of a Hermitian matrix can be identified as the product of its eigenvalues, and its trace will be their sum. Apart from the individual eigenvalues themselves, these are the most useful of the invariants that a matrix has with respect to unitary transformation.

We illustrate the ideas thus far introduced in this section in the next example.

Example 6.4.1 TRANSFORMING A MATRIX TO DIAGONAL FORM

We return to the matrix H of Example 6.2.2:

$$H = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

We note that it is Hermitian, so there exists a unitary transformation U that will diagonalize it. Since we already know the eigenvectors of H , we can use them to construct U . Noting that we need **normalized** eigenvectors, and consulting Eqs. (6.9) to (6.11), we have

$$\lambda = 2, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}; \quad \lambda = 1, \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{pmatrix}; \quad \lambda = -1, \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{pmatrix}.$$

Combining these as columns into U^{-1} ,

$$U^{-1} = \begin{pmatrix} 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1 & 0 & 0 \end{pmatrix}.$$

Since $U = (U^{-1})^\dagger$, we easily form

$$U = \begin{pmatrix} 0 & 0 & 1 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \end{pmatrix} \quad \text{and} \quad UHU^{-1} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

The trace of H is 2, as is the sum of the eigenvalues; $\det(H)$ is -2 , equal to $2 \times 1 \times (-1)$. ■

Finding a Diagonalizing Transformation

As Example 6.4.1 shows, a knowledge of the eigenvectors of a Hermitian matrix H enables the direct construction of a unitary matrix U that transforms H into diagonal form. But we are interested in diagonalizing matrices for the purpose of **finding** their eigenvectors and eigenvalues, so the construction illustrated in Example 6.4.1 does not meet our present needs. Applied mathematicians (and even theoretical chemists!) have over many years

given attention to numerical methods for diagonalizing matrices of order large enough that direct, exact solution of the secular equation is not possible, and computer programs for carrying out these methods have reached a high degree of sophistication and efficiency. In varying ways, such programs involve processes that approach diagonalization via successive approximations. That is to be expected, since explicit formulas for the solution of high-degree algebraic equations (including, of course, secular equations) do not exist. To give the reader a sense of the level that has been reached in matrix diagonalization technology, we identify a computation¹ that determined some of the eigenvalues and eigenvectors of a matrix whose dimension exceeded 10^9 .

One of the older techniques for diagonalizing a matrix is due to Jacobi. It has now been supplanted by more efficient (but less transparent) methods, but we discuss it briefly here to illustrate the ideas involved. The essence of the Jacobi method is that if a Hermitian matrix H has a nonzero value of some off-diagonal h_{ij} (and thus also h_{ji}), a unitary transformation that alters only rows/columns i and j can reduce h_{ij} and h_{ji} to zero. While this transformation may cause other, previously zeroed elements to become nonzero, it can be shown that the resulting matrix is closer to being diagonal (meaning that the sum of the squared magnitudes of its off-diagonal elements has been reduced). One may therefore apply Jacobi-type transformations repeatedly to reduce individual off-diagonal elements to zero, continuing until there is no off-diagonal element larger than an acceptable tolerance. If one constructs the unitary matrix that is the product of the individual transformations, one obtains thereby the overall diagonalizing transformation. Alternatively, one can use the Jacobi method only for retrieval of the eigenvalues, after which the method presented previously can be used to obtain the eigenvectors.

Simultaneous Diagonalization

It is of interest to know whether two Hermitian matrices A and B can have a common set of eigenvectors. It turns out that this is possible if and only if they commute. The proof is simple if the eigenvectors of either A or B are nondegenerate.

Assume that $\mathbf{c}_i$ are a set of eigenvectors of both A and B with respective eigenvalues a_i and b_i . Then form, for any i ,

$$B A \mathbf{c}_i = B a_i \mathbf{c}_i = b_i a_i \mathbf{c}_i,$$

$$A B \mathbf{c}_i = A b_i \mathbf{c}_i = a_i b_i \mathbf{c}_i.$$

These equations show that $B A \mathbf{c}_i = A B \mathbf{c}_i$ for every $\mathbf{c}_i$. Since any vector $\mathbf{v}$ can be written as a linear combination of the $\mathbf{c}_i$, we find that $(B A - A B) \mathbf{v} = 0$ for all $\mathbf{v}$, which means that $B A = A B$. We have found that the existence of a common set of eigenvectors implies commutation. It remains to prove the converse, namely that commutation permits construction of a common set of eigenvectors.

For the converse, we assume that A and B commute, that $\mathbf{c}_i$ is an eigenvector of A with eigenvalue a_i , and that this eigenvector of A is nondegenerate. Then we form

$$A B \mathbf{c}_i = B A \mathbf{c}_i = B a_i \mathbf{c}_i, \quad \text{or} \quad A(B \mathbf{c}_i) = a_i(B \mathbf{c}_i).$$

¹J. Olsen, P. Jørgensen, and J. Simons, Passing the one-billion limit in full configuration-interaction calculations, *Chem. Phys. Lett.* **169**: 463 (1990).

This equation shows that $\mathbf{B}\mathbf{c}_i$ is also an eigenvector of $\mathbf{A}$ with eigenvalue a_i . Since the eigenvector of $\mathbf{A}$ was assumed nondegenerate, $\mathbf{B}\mathbf{c}_i$ must be proportional to $\mathbf{c}_i$, meaning that $\mathbf{c}_i$ is also an eigenvector of $\mathbf{B}$. This completes the proof that if $\mathbf{A}$ and $\mathbf{B}$ commute, they have common eigenvectors.

The proof of this theorem can be extended to include the case in which both operators have degenerate eigenvectors. Including that extension, we summarize by stating the general result:

Hermitian matrices have a complete set of eigenvectors in common if and only if they commute.

It may happen that we have three matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$, and that $[\mathbf{A}, \mathbf{B}] = 0$ and $[\mathbf{A}, \mathbf{C}] = 0$, but $[\mathbf{B}, \mathbf{C}] \neq 0$. In that case, which is actually quite common in atomic physics, we have a choice. We can insist upon a set of $\mathbf{c}_i$ that are simultaneous eigenvectors of $\mathbf{A}$ and $\mathbf{B}$, in which case not all the $\mathbf{c}_i$ can be eigenvectors of $\mathbf{C}$, or we can have simultaneous eigenvectors of $\mathbf{A}$ and $\mathbf{C}$, but not $\mathbf{B}$. In atomic physics these choices typically correspond to descriptions in which different angular momenta are required to have definite values.

Spectral Decomposition

Once the eigenvalues and eigenvectors of a Hermitian matrix $\mathbf{H}$ have been found, we can express $\mathbf{H}$ in terms of these quantities. Since mathematicians call the set of eigenvalues of $\mathbf{H}$ its **spectrum**, the expression we now derive for $\mathbf{H}$ is referred to as its **spectral decomposition**.

As previously noted, in the orthonormal eigenvector basis the matrix $\mathbf{H}$ will be diagonal. Then, instead of the general form for the basis expansion of an operator, $\mathbf{H}$ will be of the diagonal form

$$\mathbf{H} = \sum_{\mu} |\mathbf{c}_{\mu}\rangle \lambda_{\mu} \langle \mathbf{c}_{\mu}|, \quad \text{each } \mathbf{c}_{\mu} \text{ satisfies } \mathbf{H}\mathbf{c}_{\mu} = \lambda_{\mu}\mathbf{c}_{\mu} \text{ and } \langle \mathbf{c}_{\mu}|\mathbf{c}_{\mu}\rangle = 1. \quad (6.25)$$

This result, the **spectral decomposition** of $\mathbf{H}$, is easily checked by applying it to any eigenvector $\mathbf{c}_{\nu}$.

Another result related to the spectral decomposition of $\mathbf{H}$ can be obtained if we multiply both sides of the equation $\mathbf{H}\mathbf{c}_{\mu} = \lambda_{\mu}\mathbf{c}_{\mu}$ on the left by $\mathbf{H}$, reaching

$$\mathbf{H}^2\mathbf{c}_{\mu} = (\lambda_{\mu})^2\mathbf{c}_{\mu};$$

further applications of $\mathbf{H}$ show that all positive powers of $\mathbf{H}$ have the same eigenvectors as $\mathbf{H}$, so if $f(\mathbf{H})$ is any function of $\mathbf{H}$ that has a power-series expansion, it has spectral decomposition

$$f(\mathbf{H}) = \sum_{\mu} |\mathbf{c}_{\mu}\rangle f(\lambda_{\mu}) \langle \mathbf{c}_{\mu}|. \quad (6.26)$$

Equation (6.26) can be extended to include negative powers if $\mathbf{H}$ is nonsingular; to do so, multiply $\mathbf{H}\mathbf{c}_{\mu} = \lambda_{\mu}\mathbf{c}_{\mu}$ on the left by $\mathbf{H}^{-1}$ and rearrange, to obtain

$$\mathbf{H}^{-1}\mathbf{c}_{\mu} = \frac{1}{\lambda_{\mu}}\mathbf{c}_{\mu},$$

showing that negative powers of $\mathbf{H}$ also have the same eigenvectors as $\mathbf{H}$.

Finally, we can now easily prove the trace formula, Eq. (2.84). In the eigenvector basis,

$$\det(\exp(\mathbf{A})) = \prod_{\mu} e^{\lambda_{\mu}} = \exp\left(\sum_{\mu} \lambda_{\mu}\right) = \exp(\text{trace}(\mathbf{A})). \quad (6.27)$$

Since the determinant and trace are basis-independent, this proves the trace formula.

Expectation Values

The expectation value of a Hermitian operator H associated with the normalized function ψ was defined in Eq. (5.61) as

$$\langle H \rangle = \langle \psi | H | \psi \rangle, \quad (6.28)$$

where it was shown that if an orthonormal basis was introduced, with H then represented by a matrix H and ψ represented by a vector $\mathbf{a}$, this expectation value assumed the form

$$\langle H \rangle = \mathbf{a}^{\dagger} H \mathbf{a} = \langle \mathbf{a} | H | \mathbf{a} \rangle = \sum_{\nu\mu} a_{\nu}^* h_{\nu\mu} a_{\mu}. \quad (6.29)$$

If these quantities are expressed in the orthonormal eigenvector basis, Eq. (6.29) becomes

$$\langle H \rangle = \sum_{\mu} a_{\mu}^* \lambda_{\mu} a_{\mu} = \sum_{\mu} |a_{\mu}|^2 \lambda_{\mu}, \quad (6.30)$$

where a_{μ} is the coefficient of the eigenvector $\mathbf{c}_{\mu}$ (with eigenvalue λ_{μ}) in the expansion of ψ . We note that the expectation value is a weighted sum of the eigenvalues, with the weights nonnegative, and adding to unity because

$$\langle \mathbf{a} | \mathbf{a} \rangle = \sum_{\mu} a_{\mu}^* a_{\mu} = \sum_{\mu} |a_{\mu}|^2 = 1. \quad (6.31)$$

An obvious implication of Eq. (6.30) is that the expectation value $\langle H \rangle$ cannot be smaller than the smallest eigenvalue nor larger than the largest eigenvalue. The quantum-mechanical interpretation of this observation is that if H corresponds to a physical quantity, measurements of that quantity will yield the values λ_{μ} with relative probabilities given by $|a_{\mu}|^2$, and hence with an average value corresponding to the weighted sum, which is the expectation value.

Hermitian operators arising in physical problems often have finite smallest eigenvalues. This, in turn, means that the expectation value of the physical quantity associated with the operator has a finite lower bound. We thus have the frequently useful relation

If the algebraically smallest eigenvalue of H is finite, then, for any ψ , $\langle \psi | H | \psi \rangle$ will be greater than or equal to this eigenvalue, with the equality occurring only if ψ is an eigenfunction corresponding to the smallest eigenvalue.

Positive Definite and Singular Operators

If all the eigenvalues of an operator A are positive, it is termed **positive definite**. If and only if A is positive definite, its expectation value for any nonzero ψ , namely $\langle\psi|A|\psi\rangle$, will also be positive, since (when ψ is normalized) it must be equal to or larger than the smallest eigenvalue.

Example 6.4.2 OVERLAP MATRIX

Let S be an **overlap matrix** of elements $s_{v\mu} = \langle\chi_v|\chi_\mu\rangle$, where the χ_v are members of a linearly independent, but nonorthogonal basis. If we assume an arbitrary nonzero function ψ to be expanded in terms of the χ_v , according to $\psi = \sum_v b_v \chi_v$, the scalar product $\langle\psi|\psi\rangle$ will be given by

$$\langle\psi|\psi\rangle = \sum_{v\mu} b_v^* s_{v\mu} b_\mu,$$

which is of the form of an expectation value for the matrix S . Since $\langle\psi|\psi\rangle$ is an inherently positive quantity, we conclude that S is positive definite. ■

If, on the other hand, the rows (or the columns) of a square matrix represent linearly dependent forms, either as coefficients in a basis-set expansion or as the coefficients of a linear expression in a set of variables, the matrix will be singular, and that fact will be signaled by the presence of eigenvalues that are zero. The number of zero eigenvalues provides an indication of the extent of the linear dependence; if an $n \times n$ matrix has m zero eigenvalues, its rank will be $n - m$.

Exercises

- 6.4.1** Show that the eigenvalues of a matrix are unaltered if the matrix is transformed by a similarity transformation—a transformation that need not be unitary, but of the form given in Eq. (5.78).

This property is not limited to symmetric or Hermitian matrices. It holds for any matrix satisfying an eigenvalue equation of the type $A\mathbf{x} = \lambda\mathbf{x}$. If our matrix can be brought into diagonal form by a similarity transformation, then two immediate consequences are that:

1. The trace (sum of eigenvalues) is invariant under a similarity transformation.
2. The determinant (product of eigenvalues) is invariant under a similarity transformation.

Note. The invariance of the trace and determinant are often demonstrated by using the Cayley-Hamilton theorem, which states that a matrix satisfies its own characteristic (secular) equation.

- 6.4.2** As a converse of the theorem that Hermitian matrices have real eigenvalues and that eigenvectors corresponding to distinct eigenvalues are orthogonal, show that if

- (a) the eigenvalues of a matrix are real and
- (b) the eigenvectors satisfy $\mathbf{x}_i^\dagger \mathbf{x}_j = \delta_{ij}$,

then the matrix is Hermitian.

- 6.4.3** Show that a real matrix that is not symmetric cannot be diagonalized by an orthogonal or unitary transformation.

Hint. Assume that the nonsymmetric real matrix can be diagonalized and develop a contradiction.

- 6.4.4** The matrices representing the angular momentum components L_x , L_y , and L_z are all Hermitian. Show that the eigenvalues of $\mathbf{L}^2$, where $\mathbf{L}^2 = L_x^2 + L_y^2 + L_z^2$, are real and nonnegative.

- 6.4.5** A has eigenvalues λ_i and corresponding eigenvectors $|\mathbf{x}_i\rangle$. Show that A^{-1} has the same eigenvectors but with eigenvalues λ_i^{-1} .

- 6.4.6** A square matrix with zero determinant is labeled **singular**.

- (a) If A is singular, show that there is at least one nonzero column vector $\mathbf{v}$ such that

$$A|\mathbf{v}\rangle = 0.$$

- (b) If there is a nonzero vector $|\mathbf{v}\rangle$ such that

$$A|\mathbf{v}\rangle = 0,$$

show that A is a singular matrix. This means that if a matrix (or operator) has zero as an eigenvalue, the matrix (or operator) has no inverse and its determinant is zero.

- 6.4.7** Two Hermitian matrices A and B have the same eigenvalues. Show that A and B are related by a unitary transformation.

- 6.4.8** Find the eigenvalues and an orthonormal set of eigenvectors for each of the matrices of Exercise 2.2.12.

- 6.4.9** The unit process in the iterative matrix diagonalization procedure known as the Jacobi method is a unitary transformation that operates on rows/columns i and j of a real symmetric matrix A to make $a_{ij} = a_{ji} = 0$. If this transformation (from basis functions φ_i and φ_j to φ'_i and φ'_j) is written

$$\varphi'_i = \varphi_i \cos \theta - \varphi_j \sin \theta, \quad \varphi'_j = \varphi_i \sin \theta + \varphi_j \cos \theta,$$

- (a) Show that a_{ij} is transformed to zero if $\tan 2\theta = \frac{2a_{ij}}{a_{jj} - a_{ii}}$,
- (b) Show that $a_{\mu\nu}$ remains unchanged if neither μ nor ν is i or j ,
- (c) Find a'_{ii} and a'_{jj} and show that the trace of A is not changed by the transformation,
- (d) Find $a'_{i\mu}$ and $a'_{j\mu}$ (where μ is neither i nor j) and show that the sum of the squares of the off-diagonal elements of A is reduced by the amount $2a_{ij}^2$.

6.5 NORMAL MATRICES

Thus far the discussion has been centered on Hermitian eigenvalue problems, which we showed to have real eigenvalues and orthogonal eigenvectors, and therefore capable of being diagonalized by a unitary transformation. However, the class of matrices which can be diagonalized by a unitary transformation contains, in addition to Hermitian matrices, all other matrices that commute with their adjoints; a matrix A with this property, namely

$$[A, A^\dagger] = 0,$$

is termed **normal**.² Clearly Hermitian matrices are normal, as $H^\dagger = H$. Unitary matrices are also normal, as U commutes with its inverse. Anti-Hermitian matrices (with $A^\dagger = -A$) are also normal. And there exist normal matrices that are not in any of these categories.

To show that normal matrices can be diagonalized by a unitary transformation, it suffices to prove that their eigenvectors can form an orthonormal set, which reduces to the requirement that eigenvectors of different eigenvalues be orthogonal. The proof proceeds in two steps, of which the first is to demonstrate that a normal matrix A and its adjoint have the same eigenvectors.

Assuming $|\mathbf{x}\rangle$ to be an eigenvector of A with eigenvalue λ , we have the equation

$$(A - \lambda \mathbf{1})|\mathbf{x}\rangle = 0.$$

Multiplying this equation on its left by $\langle \mathbf{x} | (A^\dagger - \lambda^* \mathbf{1})$, we have

$$\langle \mathbf{x} | (A^\dagger - \lambda^* \mathbf{1}) (A - \lambda \mathbf{1}) |\mathbf{x}\rangle = 0,$$

after which we use the normal property to interchange the two parenthesized quantities, bringing us to

$$\langle \mathbf{x} | (A - \lambda \mathbf{1}) (A^\dagger - \lambda^* \mathbf{1}) |\mathbf{x}\rangle = 0.$$

Moving the first parenthesized quantity into the left half-bracket, we have

$$\langle (A^\dagger - \lambda^* \mathbf{1}) \mathbf{x} | (A^\dagger - \lambda^* \mathbf{1}) \mathbf{x} \rangle = 0,$$

which we identify as a scalar product of the form $\langle f | f \rangle$. The only way this scalar product can vanish is if

$$(A^\dagger - \lambda^* \mathbf{1})|\mathbf{x}\rangle = 0,$$

showing that $|\mathbf{x}\rangle$ is an eigenvector of $A^\dagger$ in addition to being an eigenvector of A . However, the eigenvalues of A and $A^\dagger$ are complex conjugates; for general normal matrices λ need not be real.

A demonstration that the eigenvectors are orthogonal proceeds along the same lines as for Hermitian matrices. Letting $|\mathbf{x}_i\rangle$ and $|\mathbf{x}_j\rangle$ be two eigenvectors (of both A and $A^\dagger$), we form

$$\langle \mathbf{x}_j | A | \mathbf{x}_i \rangle = \lambda_i \langle \mathbf{x}_j | \mathbf{x}_i \rangle, \quad \langle \mathbf{x}_i | A^\dagger | \mathbf{x}_j \rangle = \lambda_j^* \langle \mathbf{x}_i | \mathbf{x}_j \rangle. \quad (6.32)$$

²Normal matrices are the largest class of matrices that can be diagonalized by unitary transformations. For an extensive discussion of normal matrices, see P. A. Macklin, Normal matrices for physicists, *Am. J. Phys.* **52**: 513 (1984).

We now take the complex conjugate of the second of these equations, noting that $\langle \mathbf{x}_i | \mathbf{x}_j \rangle^* = \langle \mathbf{x}_j | \mathbf{x}_i \rangle$. To form the complex conjugate of $\langle \mathbf{x}_i | \mathbf{A}^\dagger | \mathbf{x}_j \rangle$, we convert it first to $\langle \mathbf{A} \mathbf{x}_i | \mathbf{x}_j \rangle$ and then interchange the two half-brackets. Equations (6.32) then become

$$\langle \mathbf{x}_j | \mathbf{A} | \mathbf{x}_i \rangle = \lambda_i \langle \mathbf{x}_j | \mathbf{x}_i \rangle, \quad \langle \mathbf{x}_j | \mathbf{A} | \mathbf{x}_i \rangle = \lambda_j \langle \mathbf{x}_j | \mathbf{x}_i \rangle. \quad (6.33)$$

These equations indicate that if $\lambda_i \neq \lambda_j$, we must have $\langle \mathbf{x}_j | \mathbf{x}_i \rangle = 0$, thus proving orthogonality.

The fact that the eigenvalues of a normal matrix $\mathbf{A}^\dagger$ are complex conjugates of the eigenvalues of $\mathbf{A}$ enables us to conclude that

- the eigenvalues of an anti-Hermitian matrix are pure imaginary (because $\mathbf{A}^\dagger = -\mathbf{A}$, $\lambda^* = -\lambda$), and
- the eigenvalues of a unitary matrix are of unit magnitude (because $\lambda^* = 1/\lambda$, equivalent to $\lambda^* \lambda = 1$).

Example 6.5.1 A NORMAL EIGENSYSTEM

Consider the unitary matrix

$$\mathbf{U} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

This matrix describes a rotational transformation in which $z \rightarrow x$, $x \rightarrow y$, and $y \rightarrow z$. Because it is unitary, it is also normal, and we may find its eigenvalues from the secular equation

$$\det(\mathbf{U} - \lambda \mathbf{1}) = \begin{vmatrix} -\lambda & 0 & 1 \\ 1 & -\lambda & 0 \\ 0 & 1 & -\lambda \end{vmatrix} = -\lambda^3 + 1 = 0,$$

which has solutions $\lambda = 1$, ω , and ω^* , where $\omega = e^{2\pi i/3}$. (Note that $\omega^3 = 1$, so $\omega^* = 1/\omega = \omega^2$.) Because $\mathbf{U}$ is real, unitary and describes a rotation, its eigenvalues must fall on the unit circle, their sum (the trace) must be real, and their product (the determinant) must be +1. This means that one of the eigenvalues must be +1, and the remaining two may be real (both +1 or both -1) or form a complex conjugate pair. We see that the eigenvalues we have found satisfy these criteria. The trace of $\mathbf{U}$ is zero, as is the sum $1 + \omega + \omega^*$ (this may be verified graphically; see Fig. 6.3).

Proceeding to the eigenvectors, substitution into the equation

$$(\mathbf{U} - \lambda \mathbf{1}) \mathbf{c} = 0$$

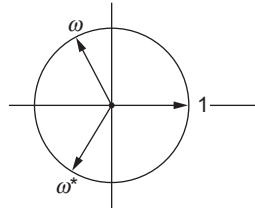


FIGURE 6.3 Eigenvalues of the matrix U , Example 6.5.1.

yields (in unnormalized form)

$$\lambda_1 = 1, \quad \mathbf{c}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \lambda_2 = \omega, \quad \mathbf{c}_2 = \begin{pmatrix} 1 \\ \omega^* \\ \omega \end{pmatrix}, \quad \lambda_3 = \omega^2, \quad \mathbf{c}_3 = \begin{pmatrix} 1 \\ \omega \\ \omega^* \end{pmatrix}.$$

The interpretation of this result is interesting. The eigenvector $\mathbf{c}_1$ is unchanged by U (application of U multiplies it by unity), so it must lie in the direction of the axis of the rotation described by U . The other two eigenvectors are complex linear combinations of the coordinates that are invariant in “direction,” but not in phase under application of U . We write “direction” in quotes, since the complex coefficients in the eigenvectors cause them not to identify directions in physical space. Nevertheless, they do form quantities that are invariant except for multiplication by the eigenvalue (which we identify as a phase, since it is of magnitude unity). The argument of ω , $2\pi/3$, identifies the amount of the rotation about the $\mathbf{c}_1$ axis. Coming back to physical reality, we note that we have found that U corresponds to a rotation of amount $2\pi/3$ about an axis in the $(1,1,1)$ direction; the reader can verify that this indeed takes x into y , y into z , and z into x .

Because U is normal, its eigenvectors must be orthogonal. Since we now have complex quantities, in order to check this we must compute the scalar product of two vectors $\mathbf{a}$ and $\mathbf{b}$ from the formula $\mathbf{a}^\dagger \mathbf{b}$. Our eigenvectors pass this test.

Finally, let’s verify that U and $U^\dagger$ have the same eigenvectors, and that corresponding eigenvalues are complex conjugates. Taking the adjoint of U , we have

$$U^\dagger = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

Using the eigenvectors we have already found to form $U^\dagger \mathbf{c}_i$, the verification is easily established. We illustrate with $\mathbf{c}_2$:

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ \omega^* \\ \omega \end{pmatrix} = \begin{pmatrix} \omega^* \\ \omega \\ 1 \end{pmatrix} = \omega^* \begin{pmatrix} 1 \\ \omega^* \\ \omega \end{pmatrix},$$

as required. ■

Nonnormal Matrices

Matrices that are not even normal sometimes enter problems of importance in physics. Such a matrix, A , still has the property that the eigenvalues of $A^\dagger$ are the complex conjugates of the eigenvalues of A , because $\det(A^\dagger) = [\det(A)]^*$, so

$$\det(A - \lambda \mathbf{1}) = 0 \quad \longrightarrow \quad \det(A^\dagger - \lambda^* \mathbf{1}) = 0,$$

for the same λ , but it is no longer true that the eigenvectors are orthogonal or that A and $A^\dagger$ have common eigenvectors.

Here is an example arising from the analysis of vibrations in mechanical systems. We consider the vibrations of a classical model of the CO_2 molecule. Even though the model is classical, it is a good representation of the actual quantum-mechanical system, as to good approximation the nuclei execute small (classical) oscillations in the Hooke's-law potential generated by the electron distribution. This problem is an illustration of the application of matrix techniques to a problem that does not start as a matrix problem. It also provides an example of the eigenvalues and eigenvectors of an asymmetric real matrix.

Example 6.5.2 NORMAL MODES

Consider three masses on the x -axis joined by springs as shown in Fig. 6.4. The spring forces are assumed to be linear in the displacements from equilibrium (small displacements, Hooke's law), and the masses are constrained to stay on the x -axis.

Using a different coordinate for the displacement of each mass from its equilibrium position, Newton's second law yields the set of equations

$$\begin{aligned}\ddot{x}_1 &= -\frac{k}{M}(x_1 - x_2) \\ \ddot{x}_2 &= -\frac{k}{m}(x_2 - x_1) - \frac{k}{m}(x_2 - x_3) \\ \ddot{x}_3 &= -\frac{k}{M}(x_3 - x_2),\end{aligned}\tag{6.34}$$

where $\ddot{x}$ stands for d^2x/dt^2 . We seek the frequencies, ω , such that all the masses vibrate at the same frequency. These are called the **normal** modes of vibration,³ and are solutions to Eqs. (6.34) with

$$x_i(t) = x_i \sin \omega t, \quad i = 1, 2, 3.$$

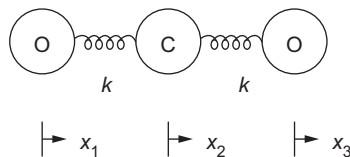


FIGURE 6.4 The three-mass spring system representing the CO_2 molecule.

³For detailed discussion of normal modes of vibration, see E. B. Wilson, Jr., J. C. Decius, and P. C. Cross, *Molecular Vibrations—The Theory of Infrared and Raman Vibrational Spectra*. New York: Dover (1980).

Substituting this solution set into Eqs. (6.34), these equations, after cancellation of the common factor $\sin \omega t$, become equivalent to the matrix equation

$$\mathbf{A}\mathbf{x} \equiv \begin{pmatrix} \frac{k}{M} & -\frac{k}{M} & 0 \\ -\frac{k}{m} & \frac{2k}{m} & -\frac{k}{m} \\ 0 & -\frac{k}{M} & \frac{k}{M} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = +\omega^2 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}. \quad (6.35)$$

We can find the eigenvalues of $\mathbf{A}$ by solving the secular equation

$$\begin{vmatrix} \frac{k}{M} - \omega^2 & -\frac{k}{M} & 0 \\ -\frac{k}{m} & \frac{2k}{m} - \omega^2 & -\frac{k}{m} \\ 0 & -\frac{k}{M} & \frac{k}{M} - \omega^2 \end{vmatrix} = 0, \quad (6.36)$$

which expands to

$$\omega^2 \left(\frac{k}{M} - \omega^2 \right) \left(\omega^2 - \frac{2k}{m} - \frac{k}{M} \right) = 0.$$

The eigenvalues are

$$\omega^2 = 0, \quad \frac{k}{M}, \quad \frac{k}{M} + \frac{2k}{m}.$$

For $\omega^2 = 0$, substitution back into Eq. (6.35) yields

$$x_1 - x_2 = 0, \quad -x_1 + 2x_2 - x_3 = 0, \quad -x_2 + x_3 = 0,$$

which corresponds to $x_1 = x_2 = x_3$. This describes pure translation with no relative motion of the masses and no vibration.

For $\omega^2 = k/M$, Eq. (6.35) yields

$$x_1 = -x_3, \quad x_2 = 0.$$

The two outer masses are moving in opposite directions. The central mass is stationary. In CO_2 this is called the **symmetric stretching** mode.

Finally, for $\omega^2 = k/M + 2k/m$, the eigenvector components are

$$x_1 = x_3, \quad x_2 = -\frac{2M}{m} x_1.$$

In this **antisymmetric stretching** mode, the two outer masses are moving, together, in a direction opposite to that of the central mass, so one CO bond stretches while the other contracts the same amount. In both of these stretching modes, the net momentum of the motion is zero.

Any displacement of the three masses along the x -axis can be described as a linear combination of these three types of motion: translation plus two forms of vibration.

The matrix A of Eq. (6.35) is not **normal**; the reader can check that $AA^\dagger \neq A^\dagger A$. As a result, the eigenvectors we have found are not orthogonal, as is obvious by examination of the unnormalized eigenvectors:

$$\omega^2 = 0, \quad \mathbf{x} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \omega^2 = \frac{k}{M}, \quad \mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad \omega^2 = \frac{k}{M} + \frac{2k}{m}, \quad \mathbf{x} = \begin{pmatrix} 1 \\ -2M/m \\ 1 \end{pmatrix}.$$

Using the same λ values, we can solve the simultaneous equations

$$(A^\dagger - \lambda^* \mathbf{1})\mathbf{y} = 0.$$

The resulting eigenvectors are

$$\omega^2 = 0, \quad \mathbf{x} = \begin{pmatrix} 1 \\ m/M \\ 1 \end{pmatrix}, \quad \omega^2 = \frac{k}{M}, \quad \mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \omega^2 = \frac{k}{M} + \frac{2k}{m}, \quad \mathbf{x} = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}.$$

These vectors are neither orthogonal nor the same as the eigenvectors of A . ■

Defective Matrices

If a matrix is not normal, it may not even have a full complement of eigenvectors. Such matrices are termed **defective**. By the fundamental theorem of algebra, a matrix of dimension N will have N eigenvalues (when their multiplicity is taken into account). It can also be shown that any matrix will have at least one eigenvector corresponding to each of its distinct eigenvalues. But it is **not** always true that that an eigenvalue of multiplicity $k > 1$ will have k eigenvectors. We give as a simple example a matrix with the doubly degenerate eigenvalue $\lambda = 1$:

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{has only the single eigenvector} \quad \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Exercises

- 6.5.1** Find the eigenvalues and corresponding eigenvectors for

$$\begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}.$$

Note that the eigenvectors are **not** orthogonal.

$$\text{ANS.} \quad \lambda_1 = 0, \quad \mathbf{c}_1 = (2, -1); \\ \lambda_2 = 4, \quad \mathbf{c}_2 = (2, 1).$$

- 6.5.2** If A is a 2×2 matrix, show that its eigenvalues λ satisfy the secular equation

$$\lambda^2 - \lambda \text{trace}(A) + \det(A) = 0.$$

- 6.5.3** Assuming a unitary matrix U to satisfy an eigenvalue equation $U\mathbf{c} = \lambda\mathbf{c}$, show that the eigenvalues of the unitary matrix have unit magnitude. This same result holds for real orthogonal matrices.

6.5.4 Since an orthogonal matrix describing a rotation in real 3-D space is a special case of a unitary matrix, such an orthogonal matrix can be diagonalized by a unitary transformation.

- (a) Show that the sum of the three eigenvalues is $1 + 2 \cos \varphi$, where φ is the net angle of rotation about a single fixed axis.
- (b) Given that one eigenvalue is 1, show that the other two eigenvalues must be $e^{i\varphi}$ and $e^{-i\varphi}$.

Our orthogonal rotation matrix (real elements) has complex eigenvalues.

6.5.5 A is an n th-order Hermitian matrix with orthonormal eigenvectors $|\mathbf{x}_i\rangle$ and real eigenvalues $\lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \dots \leq \lambda_n$. Show that for a unit magnitude vector $|\mathbf{y}\rangle$,

$$\lambda_1 \leq \langle \mathbf{y} | \mathbf{A} | \mathbf{y} \rangle \leq \lambda_n.$$

6.5.6 A particular matrix is both Hermitian and unitary. Show that its eigenvalues are all ± 1 .

Note. The Pauli and Dirac matrices are specific examples.

6.5.7 For his relativistic electron theory Dirac required a set of **four** anticommuting matrices. Assume that these matrices are to be Hermitian and unitary. If these are $n \times n$ matrices, show that n must be even. With 2×2 matrices inadequate (why?), this demonstrates that the smallest possible matrices forming a set of four anticommuting, Hermitian, unitary matrices are 4×4 .

6.5.8 A is a normal matrix with eigenvalues λ_n and orthonormal eigenvectors $|\mathbf{x}_n\rangle$. Show that A may be written as

$$\mathbf{A} = \sum_n \lambda_n |\mathbf{x}_n\rangle \langle \mathbf{x}_n|.$$

Hint. Show that both this eigenvector form of A and the original A give the same result acting on an arbitrary vector $|\mathbf{y}\rangle$.

6.5.9 A has eigenvalues 1 and -1 and corresponding eigenvectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Construct A.

$$\text{ANS. } \mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

6.5.10 A non-Hermitian matrix A has eigenvalues λ_i and corresponding eigenvectors $|\mathbf{u}_i\rangle$. The adjoint matrix $\mathbf{A}^\dagger$ has the same set of eigenvalues but **different** corresponding eigenvectors, $|\mathbf{v}_i\rangle$. Show that the eigenvectors form a **biorthogonal** set in the sense that

$$\langle \mathbf{v}_i | \mathbf{u}_j \rangle = 0 \quad \text{for } \lambda_i^* \neq \lambda_j.$$

6.5.11 You are given a pair of equations:

$$\mathbf{A} |\mathbf{f}_n\rangle = \lambda_n |\mathbf{g}_n\rangle$$

$$\tilde{\mathbf{A}} |\mathbf{g}_n\rangle = \lambda_n |\mathbf{f}_n\rangle \quad \text{with } \mathbf{A} \text{ real.}$$

- (a) Prove that $|\mathbf{f}_n\rangle$ is an eigenvector of $(\tilde{A}A)$ with eigenvalue λ_n^2 .
- (b) Prove that $|\mathbf{g}_n\rangle$ is an eigenvector of $(A\tilde{A})$ with eigenvalue λ_n^2 .
- (c) State how you know that
 - (1) The $|\mathbf{f}_n\rangle$ form an orthogonal set.
 - (2) The $|\mathbf{g}_n\rangle$ form an orthogonal set.
 - (3) λ_n^2 is real.

6.5.12 Prove that A of the preceding exercise may be written as

$$A = \sum_n \lambda_n |\mathbf{g}_n\rangle \langle \mathbf{f}_n|,$$

with the $|\mathbf{g}_n\rangle$ and $\langle \mathbf{f}_n|$ normalized to unity.

Hint. Expand an arbitrary vector as a linear combination of $|\mathbf{f}_n\rangle$.

6.5.13 Given

$$A = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & 2 \\ 1 & -4 \end{pmatrix},$$

- (a) Construct the transpose $\tilde{A}$ and the symmetric forms $\tilde{A}A$ and $A\tilde{A}$.
- (b) From $A\tilde{A}|\mathbf{g}_n\rangle = \lambda_n^2|\mathbf{g}_n\rangle$, find λ_n and $|\mathbf{g}_n\rangle$. Normalize the $|\mathbf{g}_n\rangle$.
- (c) From $\tilde{A}A|\mathbf{f}_n\rangle = \lambda_n^2|\mathbf{f}_n\rangle$, find λ_n [same as (b)] and $|\mathbf{f}_n\rangle$. Normalize the $|\mathbf{f}_n\rangle$.
- (d) Verify that $A|\mathbf{f}_n\rangle = \lambda_n|\mathbf{g}_n\rangle$ and $\tilde{A}|\mathbf{g}_n\rangle = \lambda_n|\mathbf{f}_n\rangle$.
- (e) Verify that $A = \sum_n \lambda_n |\mathbf{g}_n\rangle \langle \mathbf{f}_n|$.

6.5.14 Given the eigenvalues $\lambda_1 = 1$, $\lambda_2 = -1$ and the corresponding eigenvectors

$$|\mathbf{f}_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\mathbf{g}_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |\mathbf{f}_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \text{and}$$

$$|\mathbf{g}_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

- (a) construct A ;
- (b) verify that $A|\mathbf{f}_n\rangle = \lambda_n|\mathbf{g}_n\rangle$;
- (c) verify that $\tilde{A}|\mathbf{g}_n\rangle = \lambda_n|\mathbf{f}_n\rangle$.

$$ANS. \quad A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

6.5.15 Two matrices U and H are related by

$$U = e^{iaH},$$

with a real.

- (a) If H is Hermitian, show that U is unitary.
- (b) If U is unitary, show that H is Hermitian. (H is independent of a .)
- (c) If $\text{trace } H = 0$, show that $\det U = +1$.
- (d) If $\det U = +1$, show that $\text{trace } H = 0$.

Hint. H may be diagonalized by a similarity transformation. Then U is also diagonal. The corresponding eigenvalues are given by $u_j = \exp(iah_j)$.

6.5.16 An $n \times n$ matrix A has n eigenvalues A_i . If $B = e^A$, show that B has the same eigenvectors as A with the corresponding eigenvalues B_i given by $B_i = \exp(A_i)$.

6.5.17 A matrix P is a projection operator satisfying the condition

$$P^2 = P.$$

Show that the corresponding eigenvalues $(\rho^2)_\lambda$ and ρ_λ satisfy the relation

$$(\rho^2)_\lambda = (\rho_\lambda)^2 = \rho_\lambda.$$

This means that the eigenvalues of P are 0 and 1.

6.5.18 In the **matrix** eigenvector-eigenvalue equation

$$A|\mathbf{x}_i\rangle = \lambda_i|\mathbf{x}_i\rangle,$$

A is an $n \times n$ Hermitian matrix. For simplicity assume that its n real eigenvalues are distinct, λ_1 being the largest. If $|\mathbf{x}\rangle$ is an approximation to $|\mathbf{x}_1\rangle$,

$$|\mathbf{x}\rangle = |\mathbf{x}_1\rangle + \sum_{i=2}^n \delta_i |\mathbf{x}_i\rangle,$$

show that

$$\frac{\langle \mathbf{x} | A | \mathbf{x} \rangle}{\langle \mathbf{x} | \mathbf{x} \rangle} \leq \lambda_1$$

and that the error in λ_1 is of the order $|\delta_i|^2$. Take $|\delta_i| \ll 1$.

Hint. The n vectors $|\mathbf{x}_i\rangle$ form a **complete** orthogonal set spanning the n -dimensional (complex) space.

6.5.19 Two equal masses are connected to each other and to walls by springs as shown in Fig. 6.5. The masses are constrained to stay on a horizontal line.

- (a) Set up the Newtonian acceleration equation for each mass.
- (b) Solve the secular equation for the eigenvectors.
- (c) Determine the eigenvectors and thus the normal modes of motion.

6.5.20 Given a normal matrix A with eigenvalues λ_j , show that $A^\dagger$ has eigenvalues λ_j^* , its real part $(A + A^\dagger)/2$ has eigenvalues $\Re(\lambda_j)$, and its imaginary part $(A - A^\dagger)/2i$ has eigenvalues $\Im(\lambda_j)$.

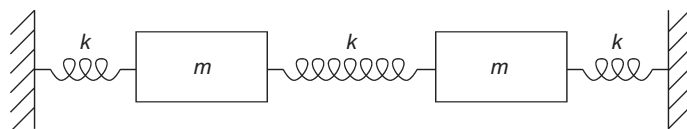


FIGURE 6.5 Triple oscillator.

6.5.21 Consider a rotation given by Euler angles $\alpha = \pi/4$, $\beta = \pi/2$, $\gamma = 5\pi/4$.

- (a) Using the formula of Eq. (3.37), construct the matrix U representing this rotation.
- (b) Find the eigenvalues and eigenvectors of U , and from them describe this rotation by specifying a single rotation axis and an angle of rotation about that axis.

Note. This technique provides a representation of rotations alternative to the Euler angles.

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